# Harbour: bounded joint-order search

CITS3011 Intelligent Agent Project 2026 | Group 55

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AI assistance is disclosed in the separate usage record.

## 1. Problem and design rationale

Harbour plays standard-map No-Press Diplomacy using the supplied Python interface. The objective is to win by reaching 18 supply centres (SCs), or retain as many as possible before the runner stops at 1920. The board is observable, but the other six powers submit simultaneous, unknown orders. Consequently, a shortest route alone is not a reliable plan: occupation, support and competing orders affect whether a move succeeds.

The design combines a utility-based agent [1], a bounded beam search [2] and a cheap evaluation function [3]. A conventional alternating two-player minimax model would misrepresent this setting. Exhaustive joint search across all seven players is also incompatible with a one-second action budget. Harbour instead searches its own current-phase joint orders and scores approximate outcomes. It is deliberately a single-phase planner, not full minimax, A*, Expectimax or MCTS.

## 2. Basic technique: heuristic beam search

For armies and fleets separately, breadth-first traversal precomputes shortest movement distances. Fleet nodes retain split coasts; an SC distance uses the nearest reachable coast without creating a coast-to-coast shortcut. Ignoring occupancy is a relaxed navigation problem [2], not a claim that the complete utility is an admissible heuristic. An unreachable target receives distance 99.

Let T be SCs not owned by this power, d(u,x,t) the relaxed distance from location x to t, and $f(u,x,t)=1/(1+d)^2$. The positional value is $V(u,x)=10\ I[x \text{ in } T]+5(f_1+f_2)$, where $f_1$ and $f_2$ are the largest target features. This values immediate capture more than travel, while retaining an incentive to advance. Unowned SCs remain targets until autumn ownership changes.

For each unit, the basic method considers hold and up to seven direct moves ranked by V. A width-10 beam progressively fixes one unit order; the remaining units temporarily hold. Joint scoring handles duplicate destinations, stationary friendly blockers and direct two-unit swaps. It interpolates current and intended position values using a rough move-success estimate, then subtracts a movement cost and a failed-move penalty. All versions share the same legal-order enumeration, retreat and adjustment routines.

**Implementation:** `new_game`, `_distance`, `_value`, `_score` and the beam section of `_movement` in `agent_55.py`. The retained `base` variant has all three improvements disabled.

## 3. Three distinct improvements

**G - global target coverage.** Independent unit rewards can repeatedly favour the same nearby SC. G halves the independent positional term and adds 9 times the sum, over targets, of the largest expected f among friendly units. A second unit approaching an already well-covered target therefore produces little extra coverage reward. This changes the evaluation structure, rather than merely tuning a weight. Its intended benefit is broader expansion; its possible cost is spreading forces when a concentrated attack is necessary. Implementation: `_features` and the `self.coverage` branches in `_score`.

**C - atomic coordination bundles.** Moving against an occupied province without support often fails, while selecting support alone may look worthless. C evaluates an attack together with one or two legal supporting orders as a single modification. Hold-support bundles are also considered. For a legal army convoy, breadth-first traversal finds a route through friendly fleets with matching convoy orders and inserts the army and fleet orders together. This changes the search neighbourhood so that complementary actions can be selected jointly.

Starting from the best beam plan, up to three best-improvement rounds compare these bundles. Support contributes only when it matches the supported unit's chosen action. Bundles use the engine's legal-order lists; the score is still approximate and does not reimplement the full adjudicator. Implementation: `_convoy_route`, bundle generation and local improvement in `_movement`, and matching-support scoring in `_score`.

**R - adaptive mobility and defence risk.** A fixed assumption about enemy holding is poor when opponents behave differently. Each power starts with smoothed move/non-move counts a=b=2. After a movement phase, counts update as a=2+0.85(a-2)+m and b=2+0.85(b-2)+(n-m), where m of n units submitted move orders. The estimated move propensity is a/(a+b). Missing orders count as holds. Discounting allows changed behaviour to influence the estimate; it is a heuristic frequency model, not a fitted Bayesian network.

Potential enemy destinations receive relative weight 4 for SCs not owned by that enemy and 1 otherwise. These weights distribute the move propensity over legal direct destinations. R uses the resulting pressure estimates to discount contested moves and vulnerable supports, penalize exposed units, and penalize unprotected owned SCs more heavily in autumn. It uses only past published orders and current legal state, not opponent identity or future information. This is motivated by uncertainty about non-optimal opponents [4]. Implementation: `update_game`, `_prepare` and `self.risk` branches in `_score`.

## 4. Operational constraints and verification

The agent performs no file writes, network requests, LLM calls or GPU computation. The four public methods retain one-second timeout decorators. Decision search has a 0.60-second internal deadline with a complete incumbent plan. Retreat destinations are distinct; builds use distinct home provinces and respect the build count; excess units are disbanded. The teacher's state synchronization is preserved.

Checks cover 56 initial power/variant combinations, a supported capture adjudicated by the engine, a friendly-fleet convoy route and adjudicated convoy, a split-coast restriction, builds, disbands and a real retreat. Every experimental turn checks returned orders against the engine's enumeration and checks duplicate origins. These checks establish interface legality, not that every support or convoy will succeed against simultaneous opposition. Synthetic 18- and 30-unit positions also completed below one second.

## 5. Experimental design and development results

The official `run_one_game` and baseline files are unmodified. Scenario 2 draws each of six opponents independently from Random, Attitude and Greedy with probabilities 0.2, 0.4 and 0.4, matching the supplied test pool. Eight configurations were evaluated for all seven powers over ten seeds each: 70 games per configuration, 560 total. Seed = 1000+100r+i, with r=0,...,9 and i the listed power index. Versions share initial opponent draws; different policies can still produce different subsequent random streams and states.

The score is min(SCs,18), matching the supplied scoring function; a win means at least 18 SCs. Table 1 reports sample standard deviation (SD), not standard error. Scenario 1 is deterministic here, so each configuration is tested once per power (seven games), rather than treating repeated identical games as independent observations. A prior 224-game pilot overlaps the development seeds and is excluded from this table. No model selection uses final holdout outcomes.

| Configuration | S2 mean SC ± SD | S2 wins / 70 | S1 mean SC |
|---|---|---|---|
| Basic | 11.59 ± 5.92 | 23 (32.9%) | 9.43 |
| Basic + G | 13.83 ± 5.44 | 35 (50.0%) | 12.29 |
| Basic + C | 16.37 ± 3.81 | 56 (80.0%) | 18.00 |
| Basic + R | 12.73 ± 4.93 | 26 (37.1%) | 9.43 |
| Basic + G + C | 16.19 ± 3.68 | 55 (78.6%) | 18.00 |
| Basic + G + R | 13.61 ± 4.77 | 30 (42.9%) | 13.43 |
| Basic + C + R * | 16.86 ± 3.14 | 59 (84.3%) | 18.00 |
| Basic + G + C + R | 15.84 ± 4.21 | 51 (72.9%) | 17.86 |

**Table 1.** Development comparisons. * Selected final configuration. All configurations completed without recorded exceptions or order-validation failures.

**Basic method.** Its 11.59 mean SCs and 32.9% wins in Scenario 2 show that navigation and bounded joint selection are useful. However, it wins none of the seven static games (9.43 mean SCs). Against stationary defenders it cannot create the supported attacks needed to turn proximity into capture.

**G.** Adding coverage alone raises mean SCs by 2.24 and wins from 23 to 35. However, adding it to C+R reduces mean SCs by 1.01 and wins from 59 to 51. The observed interaction is consistent with dispersion competing against concentrated supported attacks; the aggregate data do not prove that mechanism. G is retained for evaluation but disabled by default.

**C.** Coordination alone produces the largest single-addition gain: +4.79 mean SCs and 56 wins, versus 23 for the basic method. It wins all seven static games. Removing C from the all-enabled configuration reduces Scenario 2 mean SCs from 15.84 to 13.61. These comparisons support the value of selecting complementary orders together.

**R.** Risk alone gains 1.14 mean SCs. Adding R to C gains 0.49 mean SCs, but its exploratory paired-seed 95% bootstrap interval is [-0.50, 1.44] (5,000 resamples; seed 55). Thus the small incremental gain is uncertain. R slightly harms G+C, showing that more features are not uniformly better. C+R is selected on the best development mean (16.86) and wins (59/70), without claiming universal dominance.

## 6. Frozen final evaluation

After development, the constructor default was changed to `variant="no_coverage"` (C+R); the decision algorithms and weights were unchanged. The code was then frozen. Final Scenario 2 seeds 9000+100r+i provide 70 new games, balanced at ten per power. JSON metadata records the final source SHA-256, interpreter, platform and arguments. `PYTHONHASHSEED=0` was used throughout.

| Final condition | Games | Wins | Mean SC ± SD |
|---|---|---|---|
| Scenario 1 / static | 7 | 7 (100%) | 18.00 ± 0.00 |
| Scenario 2 / unseen seeds | 70 | 58 (82.9%) | 16.70 ± 3.58 |

**Table 2.** Final evaluation. Scenario 2's approximate 95% Wilson win-rate interval is 72.4% to 89.9%. The balanced-power design and shared baseline implementations limit interpretation as an arbitrary-opponent population estimate.

| Power | Wins / 10 | Mean SC |
|---|---|---|
| Austria | 8 | 15.20 |
| England | 8 | 17.40 |
| France | 10 | 18.00 |
| Germany | 10 | 18.00 |
| Italy | 6 | 14.60 |
| Russia | 10 | 18.00 |
| Turkey | 6 | 15.70 |

**Table 3.** Final Scenario 2 results by power. France, Germany and Russia win all ten cases; Italy and Turkey win six. These small per-power samples identify useful follow-up cases rather than stable country rankings.

The final Scenario 2 maximum public-method time was 0.081 s; the 95th-percentile decision time was 0.036 s. Maximum worker RSS was 92.9 MiB, including Python, all opponents and engine copies, so it is not an isolated agent-memory measurement. No validation failure or timeout was observed. These measurements are machine-specific and do not guarantee identical timing during marking.

The observed Scenario 1 and 2 results exceed their published five-point thresholds. No official Hidden Agent or class tournament was available, so Scenario 3 performance, bonus points and the final project mark are unknown. Main limitations are one-phase planning, an approximate combat model, weakly calibrated opponent probabilities and possible coordinated-order interactions. Future work should prioritize exact tactical adjudication within the time budget and evaluation against genuinely stronger unseen opponents, rather than more unvalidated heuristic terms.

**Reproduction:** install the supplied requirements, run `python test_55.py --checks`, then use the commands and raw result prefixes in the accompanying README. Development and holdout evidence is retained separately. All three modifications and the basic method remain in submitted code, including disabled G.

## References

[1] D. Liu. *CITS3011: Agents*, 2026, slides 29-31. [2] D. Liu. *Informed Search Algorithms*, 2026, slides 23-26, 31-33. [3] D. Liu. *Game Playing*, 2026, slides 22-29. [4] D. Liu. *Uncertainty*, 2026, slides 6-12. Lecture slides supplied with the assessment context. [5] *Diplomacy 1.1.2 engine API*, diplomacy.readthedocs.io/en/stable/api/diplomacy.engine.game.html (accessed 19 September 2026). Engine API only; no existing Diplomacy agent solution reused.